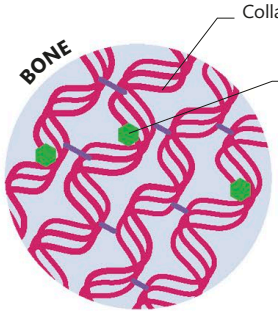
Composites

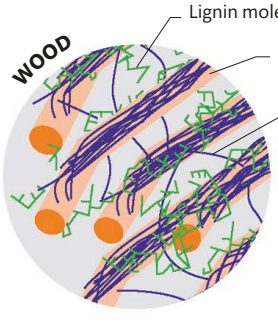
A composite material comprises two or more materials that, when combined, have superior qualities to the originals. Many modern synthetic composites are made to be strong but light.

Natural composites

Almost all the materials we see around us are composites, including many natural materials, such as wood and rock. Our bodies contain composite materials, most notably our bones and teeth, which both feature a combination of a hard outer layer and a soft inner layer. Mud bricks and the wattle-and-daub wall construction method are examples of simple combinations of basic natural materials forming composites used for their superior strength.



Structure of bone
Bone is made from a hard but brittle material called hydroxyapatite (which is mainly calcium phosphate) and a soft, flexible material called collagen (which is a protein).

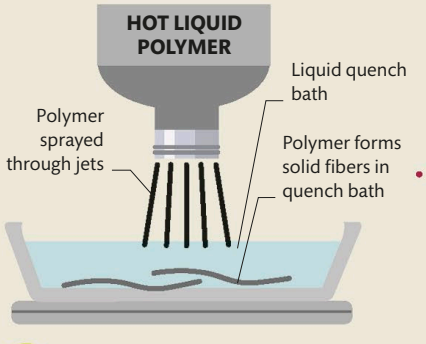


Structure of wood
Wood is made from long cellulose fibers held together by other materials. When combined, these weak materials form a strong composite.

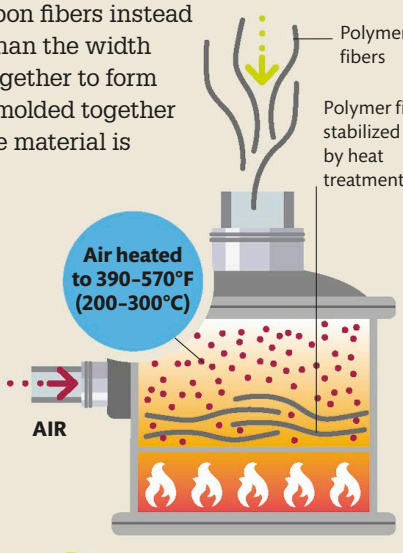
Synthetic composites

One of the first modern composites was fiberglass. It combines fine threads of glass with plastic. Advanced composites are now made out of carbon fibers instead of glass. These fibers are narrower than the width of a human hair. They are twisted together to form a yarn that is woven into cloth then molded together with a resin. The resulting composite material is extremely strong and light.

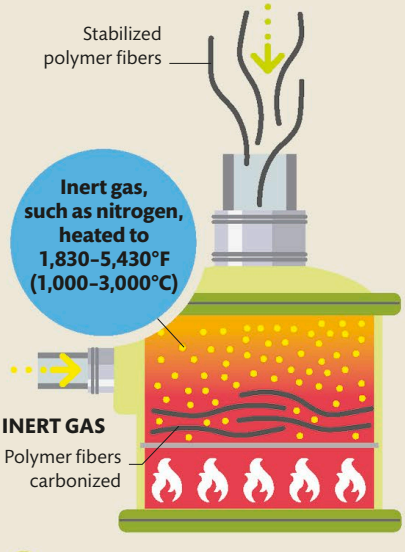
Making carbon fiber polymer
The part-chemical, part-mechanical process of making carbon fiber polymer involves a variety of gases and liquids. Exact compositions vary and are often considered trade secrets.



1 Polymer fibers produced
The raw material used to make carbon fiber is a polymer. About 90 percent of all carbon fiber is made from a polymer called polyacrylonitrile (PAN). In the first stage, the PAN is formed into long fibers.



2 Fibers stabilized
Heat chemically alters the fibers, converting their atomic bonds to a form that is more thermally stable. Oxygen molecules in the introduced air facilitate this process.



3 Fibers carbonized
The fibers are then heated to a far higher temperature in a furnace filled with an oxygen-free, inert gas to prevent the fibers from burning. As a result, the fibers lose their noncarbon atoms and become carbonized.